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# CAPSTONE PROJECT REPORT

(Project Term January-May 2026)

## An AI-Driven Smart Community Health Monitoring and Early Warning System for Water-Borne Diseases

### Submitted by

Ayush Wadatkar      Registration Number : 12222456  
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Ashrav Gurung          Registration Number : 12217793

Project Group Number : 2C509

Course Code : CSE439

### Under the Guidance of

Dr. Jaspreet Kaur

### School of Computer Science and Engineering

Lovely Professional University, Phagwara



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### TOPIC APPROVAL PERFORMA

School of Computer Science and Engineering

Program : P132::B.Tech (Computer Science and Engineering)

COURSE CODE : CSE439

REGULAR/BACKLOG : Regular

GROUP NUMBER : 2RGC0509

Supervisor Name : Dr. Jaspreet Kaur

UID : 31603

Designation : Associate Professor

Qualification :

Research Experience :

| SR.NO. | NAME OF STUDENT    | Prov. Regd. No. | BATCH | SECTION |
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| 5      | Ashrav Gurung      | 12217793        | 2022  | 322VF   |

SPECIALIZATION AREA : DevOps

Supervisor Signature: \_\_\_\_\_

PROPOSED TOPIC : Smart Community Health Monitoring and Early Warning System for Water-Borne Disease in Rural Northeast India

| Qualitative Assessment of Proposed Topic by PAC |                                                                                                                                                                                           |                    |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| Sr.No.                                          | Parameter                                                                                                                                                                                 | Rating (out of 10) |
| 1                                               | Project Novelty: Potential of the project to create new knowledge                                                                                                                         | 6.75               |
| 2                                               | Project Feasibility: Project can be timely carried out in-house with low-cost and available resources in the University by the students.                                                  | 5.75               |
| 3                                               | Project Academic Inputs: Project topic is relevant and makes extensive use of academic inputs in UG program and serves as a culminating effort for core study area of the degree program. | 6.00               |
| 4                                               | Project Supervision: Project supervisor's is technically competent to guide students, resolve any issues, and impart necessary skills.                                                    | 6.50               |
| 5                                               | Social Applicability: Project work intends to solve a practical problem.                                                                                                                  | 7.25               |
| 6                                               | Future Scope: Project has potential to become basis of future research work, publication or patent.                                                                                       | 6.75               |

| PAC Committee Members                              |            |                        |
|----------------------------------------------------|------------|------------------------|
| PAC Member (HOD/Chairperson) Name: Dr. Vikas Verma | UID: 11361 | Recommended (Y/N): Yes |
| PAC Member (Allied) Name: Dr. Harpreet Kaur        | UID: 27633 | Recommended (Y/N): Yes |
| PAC Member 3 Name: Dr. Gauri Mathur                | UID: 11400 | Recommended (Y/N): Yes |

Final Topic Approved by PAC: Smart Community Health Monitoring and Early Warning System for Water-Borne Disease in Rural Northeast India

Overall Remarks: Approved

PAC CHAIRPERSON Name: 17442::Dr. Arun Malik

Approval Date: 27 Dec 2025

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12

## DECLARATION

4 5

We hereby declare that the project work entitled An AI-Driven Smart Community Health Monitoring and Early Warning System for Water-Borne Diseases is an authentic record of our own work carried out as requirements of Capstone Project for the award of **B.Tech degree in Computer Science and Engineering** from Lovely Professional University, Phagwara, under the guidance of **Dr. Jaspreet Kaur**, during January to May 2026. All the information furnished in this capstone project report is based on our own intensive work and is genuine.

**Project Group Number:** 2C509

17

|                                              |                         |
|----------------------------------------------|-------------------------|
| <b>Name of Student 1:</b> Ayush Wadatkar     | <b>Reg No:</b> 12222456 |
| <b>Name of Student 2:</b> Ayush Jha          | <b>Reg No:</b> 12222791 |
| <b>Name of Student 3:</b> Palle Manoharshith | <b>Reg No:</b> 12223684 |
| <b>Name of Student 4:</b> Ashrav Gurung      | <b>Reg No:</b> 12217793 |

2

.....  
Signature of Student 1 Date:

.....  
Signature of Student 2 Date:

.....  
Signature of Student 3 Date:

.....  
Signature of Student 4 Date:

## CERTIFICATE

This is to certify that the declaration statement made by this group of students is correct to the best of my knowledge and belief. They have completed this Capstone Project under my guidance and supervision. The present work is the result of their original investigation, effort and study. No part of the work has ever been submitted for any other degree at any University. The Capstone Project is fit for the submission and partial fulfillment of the conditions for the award of **B.Tech in Computer Science and Engineering** from Lovely Professional University, Phagwara.

**Signature and Name of the Mentor**

**Designation:** Assistant professor

**School of Computer Science and Engineering,**  
Lovely Professional University,  
Phagwara, Punjab.

**Date: 28 April 2026**

## ACKNOWLEDGEMENT

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Finally, we would like to thank everyone who directly or indirectly contributed to the successful completion of this project.

Project Group Members

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# 1. INTRODUCTION

Water-borne illnesses like Cholera, Typhoid and Hepatitis A continue to provide public health challenges around the world. Water-borne disease incidence is particularly high in rural and resource-limited areas where clean water and proper sanitation facilities are limited. Experts have estimated significant portions of disease in developing countries occur as result of the use of tainted drinking water, as well as poor hygiene practices. Poor detection, monitoring systems or lack thereof, further compound these issues and lead to outbreaks of disease that can have tremendous public health and economic impacts.

The majority of individuals living in rural areas, including parts Northeast India, when do not have access to adequate healthcare facilities, or when water quality monitoring is done infrequently or manually, will experience delays in determining the presence of a health risk and will therefore have no way to intervene in a timely manner. Traditional disease surveillance systems rely on laboratory testing and manual reporting. Traditional methods are also cumbersome, inefficient and in remote areas are subject to long delays. Studies on mobile technology, cloud computing and artificial intelligence have demonstrated the feasibility of designing intelligent systems capable of providing real-time health monitoring and early alerts. By using modern smartphones that include advanced sensors and cameras, the collection of data in on-site locations is achievable and the use of machine learning can be integrated to help predict the likelihood of an outbreak of the disease.

The goal of the proposed system is to address these challenges through the use of integrated data from multiple sources, including; water quality parameters, visually analyzing the water sample, and symptoms reported by people who have used water, with the use of machine learning to provide predictions of potential disease outbreaks, with the capability to send out alerts in real-time to users and government agencies.

In addition, the proposed system has been designed to operate in an environment that has limited connectivity and incorporates offline capabilities and synchronizes with the cloud. A mobile app is the main user interface while there is also a centralized dashboard that allows authorities to monitor health trends and take preventative actions.

This system overall can act as a decision making support system and helps with early detection, better allocation of resources, and better health management for rural community public health.

## 2. PROFILE OF THE PROBLEM. RATIONALE/SCOPE OF THE STUDY (PROBLEM STATEMENT)

A significant risk to public health worldwide is from waterborne illnesses. Rural and developing areas of the world are particularly at risk because of poor access to clean drinking water and sanitation facilities. Waterborne diseases, including but not limited to Cholera, Typhoid and Hepatitis-A are caused by drinking water that has been contaminated and result in infection for millions of people globally. Delayed detection of waterborne bacteria through an absence of ongoing monitoring systems increases the likelihood of outbreaks of these illnesses.

In many parts of India, particularly rural areas, the assessment of the quality of drinking water is performed manually and at infrequent intervals. The delays in reporting and response to health concerns prevent the early identification of localized health issues increasing the likelihood that they will develop into an epidemic. The lack of adequate healthcare infrastructure in rural and poor regions also prevents the early identification of infected individuals and the timely treatment of those individuals.

Lastly, the absence of integrated systems that can evaluate environmental conditions, health-related symptoms, and analyse the data in real-time is also a serious hurdle to identifying potential disease outbreaks from contaminated water. Most of the regional systems are limited to laboratory based testing or require the availability of high-speed internet connection to be useful. As a result, much of the testing that is being done is becoming useless to rural areas where the ability to connect to the internet is limited.

This study will create a scalable and intelligent system to address the limitations of current methods through the use of modern technologies such as Artificial Intelligence, Machine Learning and Mobile Computing for early detection and prediction of diseases transmitted via water. The proposed system will utilize an analysis of multiple data sources such as water quality parameters, signs or symptoms, and images of water samples to accomplish this.

The proposed system will also create a Smart Community Health Monitoring and Early Warning System (SCHMEWS) capable of working efficiently with both online and offline technologies. The system is primarily targeted at rural populations, and will provide real time alerts, mobile interfaces that are easy to use, and a decision support tool for government officials. The only limitation of this smart health

monitoring system is that it makes predictions and provides an early warning, but will not replace clinical evaluations or lab testing.

Overall, this project will address the need for a cost effective, accessible, technology-based solution to improve public health surveillance of water-borne disease, and minimize the negative consequences of disease produced by contaminated water sources.

## 3. EXISTING SYSTEM

### 3.1 Introduction

Many different systems currently exist to monitor diseases transmitted through water sources. Most of them are old-fashioned and require a significant amount of manual labor to operate. In fact, many systems rely on inspecting sources of water, testing for bacteria in a laboratory, manually documenting the results, and periodically checking the quality of water.

Water quality monitoring in many rural and semi-urban areas usually occurs infrequently because of a shortage of staff trained to do the monitoring, inadequate physical infrastructure, and limited monetary resources. Many locations experience prolonged periods of time in which potential waterborne disease(s) may occur without detection, resulting in an increased risk of disease outbreaks such as Cholera, Typhoid Fever, and Hepatitis A. Without real-time monitoring and/or early warning systems to alert health officials to a potential outbreak of disease, the ability of health officials to take proactive measures to prevent the occurrence of a particular waterborne disease is seriously hampered. Consequently, the existing system focuses primarily on responding to outbreaks after they occur, a practice that is largely ineffective in preventing outbreaks.

### 3.2 Existing Software

There are a number of different systems used to gather and analyze data concerning the health of individuals, as well as to monitor the spread of disease due to the availability of software programs from both government and private entities. These systems relate information to the patients' records and hospital records, as well as the epidemiological information supporting both. These systems may take the form of large central health management systems or reporting systems that are used by the public health departments to gather, store and support the reporting of relevant health information within communities.

These systems, however, have several shortcomings related to monitoring the spread of water-borne diseases. They do not take into account relevant environmental factors, such as the quality of water, (the pH level, the turbidity, or the level of contamination); they rely heavily on having a continuous internet connection

(impractical in many remote rural areas); they rely on data collected in the past instead of collecting real-time monitoring data they do not provide users with the ability to predict occurrences of water-borne disease.

These systems are also not real user-friendly to the average citizen and are designed primarily for the benefit of the persons working for the administrators that operate the public health department. These systems do not currently provide adequate support for mobile collecting of data, for collecting images, and providing automatic alerts to personnel, which will all be critical to any new modern monitoring system.

### 3.3 DFD for Present System

At present, the **Data Flow Diagram** (DFD) represents a manual, sequential workflow governing how water samples will be collected (from wells, rivers and storage tanks), analyzed in laboratories then documented after the analyses have been completed. These analyses are then sent to the appropriate authorities/health departments.

Each analysis is then examined for the presence of contamination, and if there is evidence of contamination, the appropriate agency will take necessary actions based upon the results of the tests conducted. There is significant delay from the time the sample is transported to the lab for testing through the time the results are reported to the member of the health department (the person who will take action to remedy the contamination) due to the fact that each of these activities is performed sequentially.

In addition, the current data collection method and the data collection point to decision-maker data flow do not have an immediate notification system for that decision-maker when the water has been contaminated. For example, if a contaminant is found in a sample, the decision maker will not know to take action until the sample is completed, then it will need to go through an approval process whereby they will not know what action to take for the contaminated sample.

The current lack of real-time data collection, and lack of automated alerts leads to additional inefficiencies preventing outbreaks from being managed successfully. Current DFD analysis indicates a lack of automation, unidirectional flow of data, minimal interaction of data, and no feedback loop resulting in an overall inefficient solution for preventing contamination of drinking water.

## 4. LEVEL 0 DFD (CONTEXT DIAGRAM)

### 4.1 Description

The Level 0 DFD represents the Sahas 2.0 system as a single process interacting with external entities.

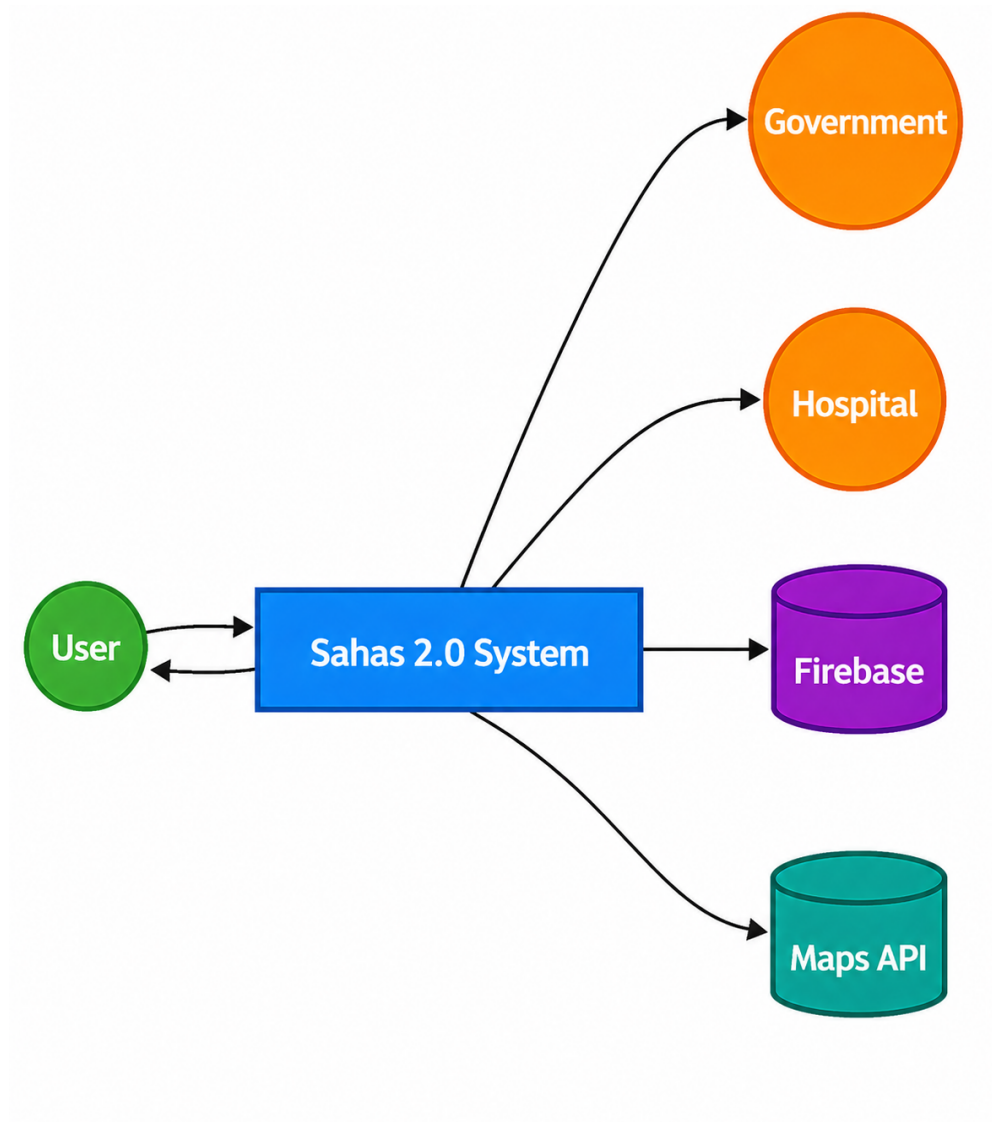
### 4.2 External Entities

- User (Citizen)
- Government Authority
- Hospital / Doctor
- Firebase Cloud
- Google Maps API

### 4.3 Data Flows

- User → System: Symptoms, water images, location data
- System → User: Health advice, results, alerts
- System → Government: Reports and analytics
- System → Hospital: Appointments and emergency data
- System → Firebase: Data storage and retrieval
- System → Maps API: Location services

#### 4.4 Diagram



### 5. LEVEL 1 DFD (MAJOR PROCESSES)

#### 5.1 Description

The Level 1 DFD decomposes the system into major functional modules.

#### 5.2 Core Processes

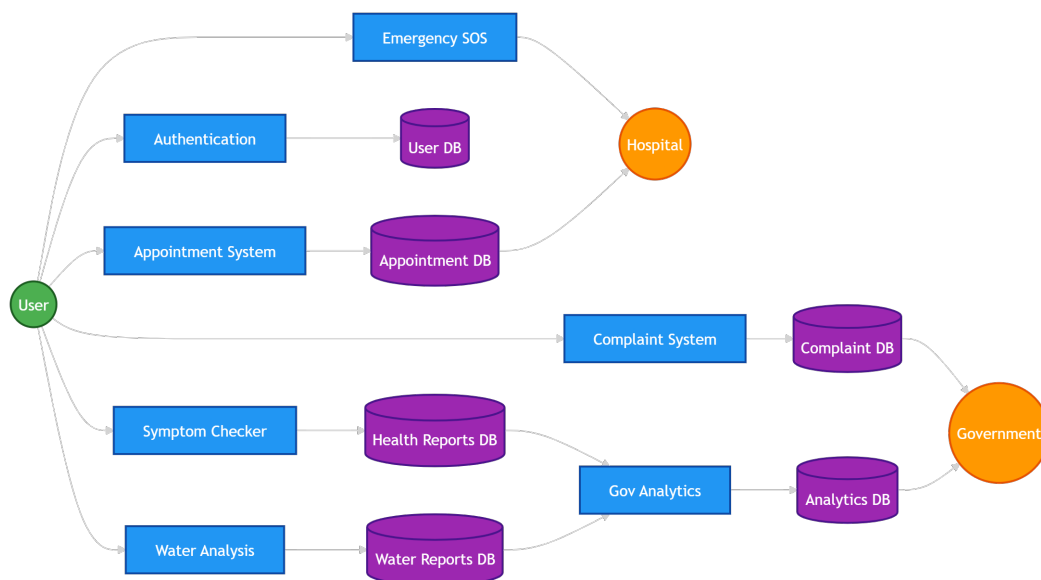
1. Authentication and Profile Management
2. Water Quality Analysis
3. Symptom Checker

4. Triage and Appointment System
5. Emergency SOS
6. Government Analytics
7. Complaint Management

### 5.3 Data Stores

- D1: User Database
- D2: Health Reports Database
- D3: Water Reports Database
- D4: Appointment Database
- D5: Complaint Database
- D6: Analytics Database

### 5.4 Diagram

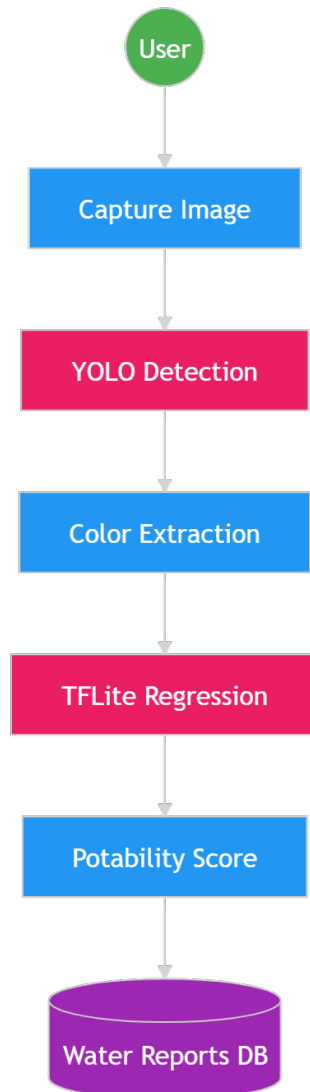


## 6. LEVEL 2 DFD (DETAILED PROCESSES)

### 6.1 Water Quality Analysis

Sub-processes:

- Image Capture
- YOLO Detection
- Color Extraction
- TFLite Regression
- Potability Score Calculation

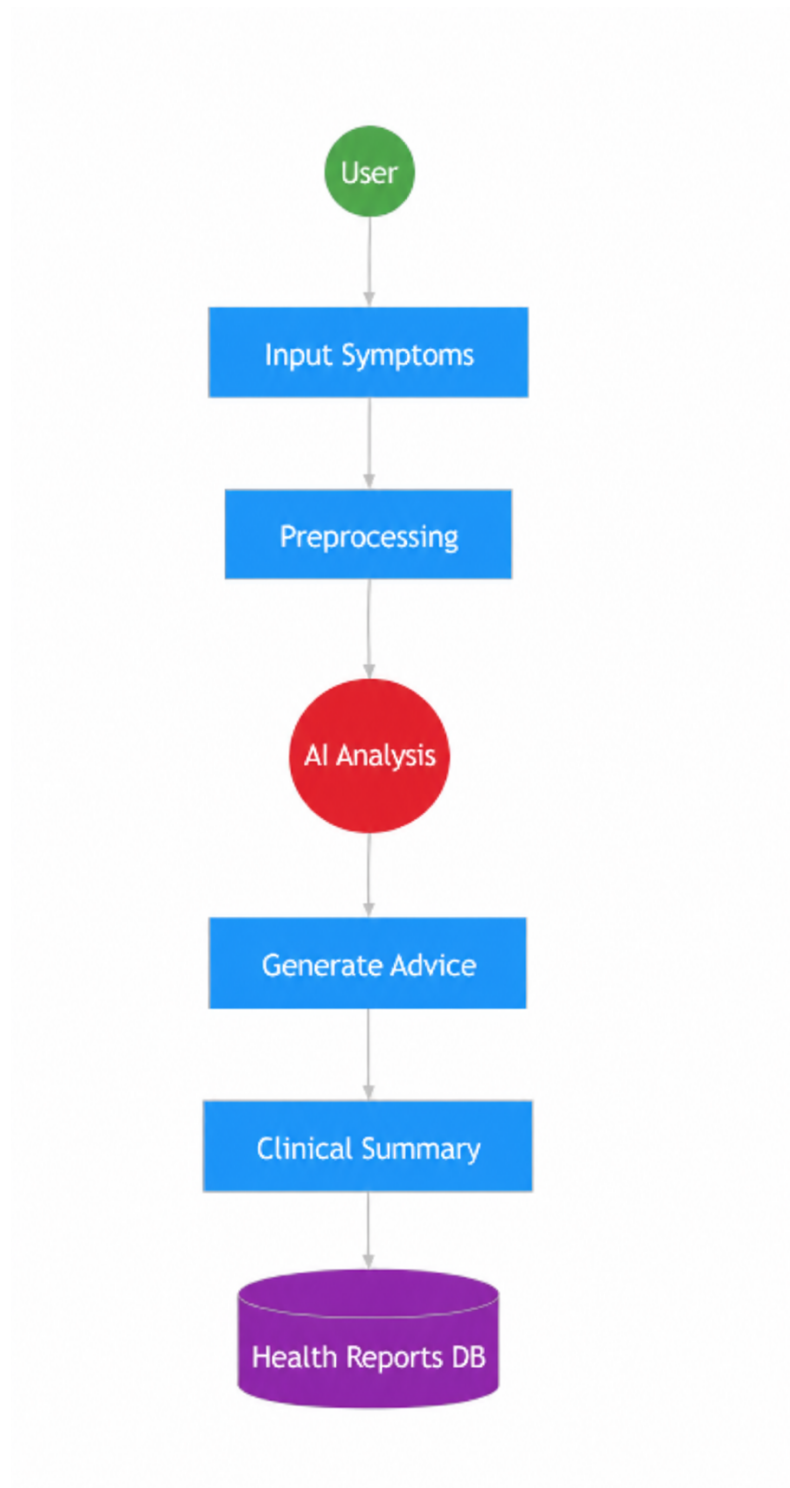


## 6.2 Symptom Checker

### Sub-processes:

- Input Symptoms
- Preprocessing

- AI Analysis
- Generate Recommendation
- Clinical Summary
- Health Report



### 6.3 Appointment System

#### Sub-processes:

- Location Fetch
- Hospital Search
- Availability Check
- Booking
- Confirmation

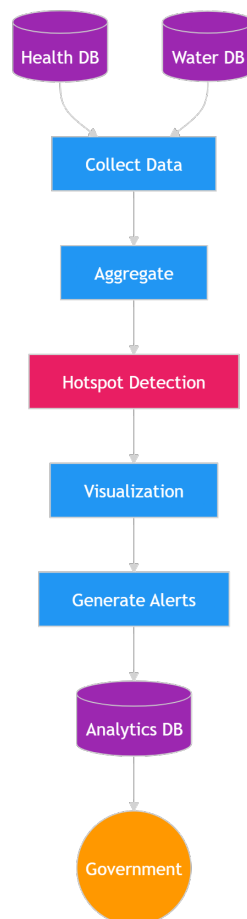
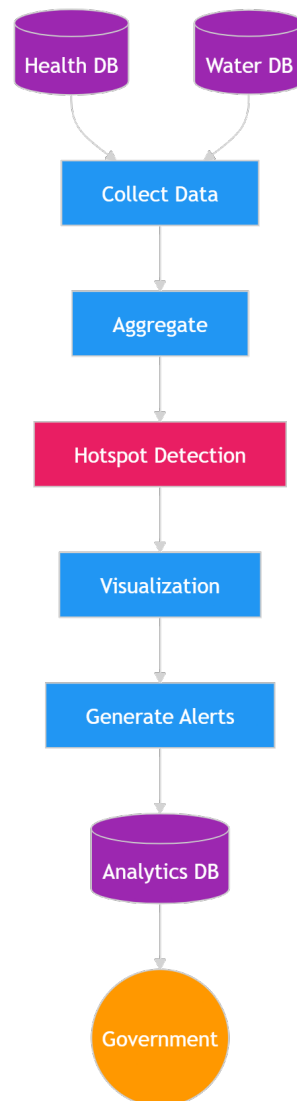


Figure 1: Triage and Appointment System DFD

## 6.4 Government Analytics

### Sub-processes:

- Data Collection
- Data Aggregation
- Hotspot Detection
- Visualization
- Alert Generation



## 6.5 What's New in the System to be Developed

The proposed system is a significant improvement over the present system because the application allows for the use of advanced technologies (Artificial Intelligence, Machine Learning and Mobile Computing). Unlike the current system where monitoring of diseases is carried out reactively, the new system will be able to monitor and predict outbreak occurrences in real-time and proactively.

The system will combine multiple types of data: the quality of water (e.g., level of contamination), "image" or image-based analyses of water samples, and health symptoms reported by people using the water. The combination of these datasets will allow the application of machine learning algorithms to analyse the datasets and identify specific patterns that are indicative of potential disease outbreak or risk, enabling early detection of outbreaks before they become widespread.

Another advantage of the proposed system is that it enables data collection and analysis when there is little or no connectivity throughout the data collection process and then synchronises with a cloud-based server later. The end-user mobile application provides a simple, intuitive, user-friendly interface for data entry and receiving alerts; while, the centralised dashboard enables authorities to view trends over time and respond accordingly.

In addition, the proposed system supports automated notification, faster data processing, and better decision-making. By enhancing the performance of the current system, the proposed solution will improve public health safety, reduce the time it takes to respond to outbreaks, and offer a scalable and efficient monitoring framework.

## 7. PROBLEM ANALYSIS

### 7.1 Product Definition

The Intelligent Health Tracking and Early Warning Systems for Water-Borne Disease, driven by AI, addresses the identification and prediction of waterborne diseases in rural and semi-urban areas. This system relies on advanced, proprietary technologies (AI/ML/Mobile Field-Based Computing) to provide ongoing real-time monitoring of health and water quality. The system continuously collects, analyzes and alerts users when there is a pattern of symptoms when users send in pictures of their water sample as their current “condition.” Any pattern found by machine learning will produce predictive analytics to predict outbreaks based on patterns identified.

This product’s purpose is to deliver a cost-effective, flexible, and simple method for identifying early signs of water-related health issues so that people can make fast choices in regards to themselves and provide support to agencies that assist the public in making choices about outbreaks of disease caused by water.

### 7.2 Feasibility Analysis

The feasibility of a proposed system is calculated in the terms of technical, economic, and operational aspects.

#### Technical Feasibility:

This system is known to be technically feasible because the technology is generally accessible when the system is implemented using readily available technologies (e.g., smartphones). Both the tools and the platforms required to develop the system (i.e., mobile development frameworks and machine learning frameworks) are also generally accessible and can adequately support the creation of the system. Additionally, since the system will function even when the connected environment is low, it meets the needs of rural communities.

#### Economic Feasibility:

This type of system can be used to implement a cost-effective solution to laboratory expenses and manual check-ups. The majority of the system involves software that

is much cheaper to develop and run than any current system available. Furthermore, open-source technologies assist in further lowering overall project costs. As a result, this system will be an affordable option for large-scale applications.

### Operational Feasibility:

The solution is so easy to use that anyone can understand it, and therefore it is available to all users, even those without any technical expertise. Users can quickly and easily enter data into the mobile application and will receive alerts regarding their submitted data. The centralized dashboard provides an easily accessible way for authorities to see all data that has been submitted from one location. The solution has been designed to be easy to implement into existing workflows, thus increasing the productivity of existing business processes while making it relatively easy for users to complete their existing processes.

## 7.3 Project Plan

By using a phased approach with a structured development process, we have achieved consistent progress through the project while also managing resources effectively and ensuring the delivery of an acceptable product on-time at each phase of the project. Each of the project phases was planned to deliver a particular outcome at the conclusion of each phase of the project through the delivery of products during each phase of the project.

### 1. Requirement Analysis Phase:

In this phase, the problem is thoroughly studied, and the requirements of the system are identified. This includes understanding user needs, analyzing existing systems, and defining functional and non-functional requirements. Data sources such as water quality parameters, symptom inputs, and image-based analysis are also identified during this stage.

### 2. System Design Phase:

The architecture of the entire system will be determined in the third phase of development. This will include the design of the data flow and how it stores its data, as well as the layout of the user interface. To aid in understanding the workflow of

the system different types of diagrams such as DFDs, system architecture and/or flowchart are created during this phase. The primary goal of the design phase is to ensure that the system is developed with an appropriate level of scalability, performance, and usability.

### **3. Development Phase:**

The next phase is the implementation of the actual system. During this phase, the mobile app will be created that will allow users to gather information be notified of any changes to their health status. The back end will also be developed to process store the information submitted by the users. In addition, a variety of machine learning algorithms [chae2018, yang2020] will be used to analyze the data from the users to forecast future health risk to the user.

### **4. Testing Phase:**

Testing is done rigorously to verify that the system is functional, reliable, and performs as expected. There are multiple forms of testing performed: unit, integration, and system. Errors and bugs found during testing will be fixed to enhance system quality.

### **5. Deployment Phase:**

Once the testing process has been completed successfully, the system will go into production (deployment). The mobile application is available to users; the backend application is typically hosted remote with either a server option or cloud-based system. Required configurations will be implemented at this time to ensure proper operation of the application.

### **6. Maintenance Phase:**

Continued Monitoring of the System for Performance Improvement and Defect Resolution (Updates) Based on Feedback from Users. The last component of this project is to continue to monitor, improve and enhance the system through the implementation of ongoing improvements using updates that represent the needs and responses of users. This means ensuring that the system continues to function at an optimal level through performance improvements and defect resolutions.

The project will be developed using a Phased Development Methodology, thus enabling a clear, efficient, and successful implementation of the system being developed.

## 8. SOFTWARE REQUIREMENT ANALYSIS

### 8.1 Introduction

Software Requirements Analysis (SRA) is a method for gathering and analyzing the requirements for the desired software application. SRA will help to clarify what the system will do and identify constraints associated with the operation of the system. The functional and non-functional requirements for the proposed AI-driven community health-monitoring solution will be contained in this section.

The purpose of the system is to monitor water quality, collect user health data, and identify users who may be at risk for water-related diseases by providing early alerts through the collection of this information or through other means (e.g., email). Improved decision-making, reduced response times and increased safety from waterborne diseases are expected outcomes from implementing this system.

### 8.2 General Description

This proposed mobile/cloud-based application will allow the general public and health officials to monitor and anticipate outbreaks of water-based diseases by entering in symptoms they've experienced, taking pictures of the contaminated water source, and receiving alerts about those hazards via this application

The application will use machine learning algorithms to gather and analyze data and give the information back to users in nearly real-time. Health officials can also analyze trends and take preventative measures via this centralized data dashboard. The application will work both online and offline, allowing for accessibility in rural areas that have limited internet access.

#### User Classes:

- **General Users:** Individuals who inputs the symptoms and receive the alerts.

- **Government Authorities:** Officials who monitors the data and manage the responses.
- **Healthcare Providers:** Professionals who analyze health trends and provide guidance based on results.

### Operating Environment:

- Android-based mobile devices
- Cloud servers for data storage and processing
- Internet connectivity (optional for offline mode)

## 8.3 Specific Requirements

### Functional Requirements:

- The system shall allow users to register and log in securely.
- The system shall allow users to input symptoms related to water-borne diseases.
- The system shall allow users to upload images of water samples.
- The system shall analyze input data using machine learning algorithms.
- The system shall predict potential disease risks based on input data.
- The system shall generate real-time alerts and notifications.
- The system shall provide a dashboard for authorities to monitor data.
- The system shall store user data securely in a database.
- The system shall support offline data collection and synchronization.

### Non-Functional Requirements:

- **Performance:** The system should provide fast and real-time responses.
- **Reliability:** The system should operate consistently without failure.
- **Usability:** The interface should be simple and user-friendly.

- **Security:** User data must be protected using authentication and encryption.
- **Scalability:** The system should handle increasing users and data efficiently.
- **Availability:** The system should be accessible whenever required.

#### System Constraints:

- Limited internet connectivity in rural areas
- Dependency on mobile device hardware capabilities
- Accuracy depends on quality of input data

## 9. DESIGN

### 9.1 System Design

The system's architecture and workflow are determined by its design, which includes a modular architecture consisting of a mobile app, a backend server, and machine learning components that all interact with each other through a unified framework.

Mobile Applications (Mobile Applications Function As Primary User Interfaces) Provide Users With The Ability To Input Symptoms, Upload Images Of Water Samples, And Receive Alerts About Their Health Status. The Backend Server Manages The Processing, Storage, And Transfer Of Information Between The Different Modules Of The System. The Collected Data Is Analyzed By Machine Learning Models To Identify Potential Disease Risks.

By Using A Modular Way Of Structuring Its Functionality, The System Provides An Effective Flow Of Information Among Its Components, As Well As The Ability To Scale Up Or Down Based On Creation Of Additional Nodes To Serve Users (e.g., Additional Mobile Applications, More Servers), And Processing Data In Real Time; Thus, Allowing Users To Access Information Either When They Have An Internet Connection (Online) Or When They Do Not (Offline) – The Mobile Application Provides Users With Offline Functionality – This Is Particularly Relevant In Rural Areas With Limited Internet Coverage.

### 9.2 Design Notations

System structure and workflow can be represented using design notations, and thus the design of the proposed SCEPT system will use design notations.

#### 1. Data Flow Diagram (DFD) Notations:

- **External Entities (Rectangles):** Indicate the place that data travels from outside of the system. Types may include: Community Member, Government Official
- **Processes (Circles/Rounded Rectangles):** Represent the transformation of data within the system, such as AI Inference and Cloud Synchronization.

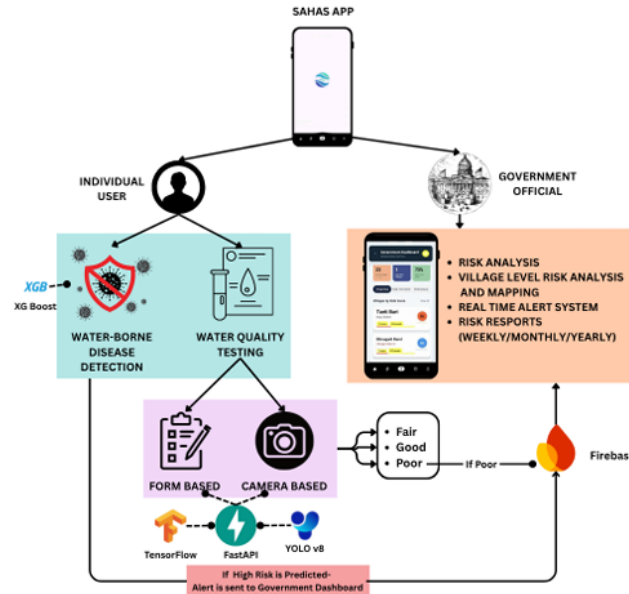


Figure 2: System Architecture

- **Data Stores (Open-ended Rectangles):** Indicate where data will be stored (examples: Firebase Firestore and Local Cache) after it has left the system.
- **Data Flows (Arrows):** Indicate how data travels from one entity to another, from entity to process, and from entity to data store.

## 2. System Architecture Notations:

- **Client Side (Mobile Application):** The client-side mobile app is a device container that contains the user interface and edge AI processing logic.
- **Backend Services (Cloud Components):** These are the cloud components of a system, such as Firebase Firestore, Authentication and Storage.
- **External APIs:** These are service blocks, such as the Google Maps API, which provide additional functionality to the application.

## 3. Flowchart Notations:

- **Start/End (Oval):** An oval-shaped symbol indicating the start and end of a process.

- **Input/Output (Parallelogram):** A parallelogram-shaped symbol used to represent the inputting of data (e.g., uploading a photograph) and outputting of data (e.g. presenting results).
- **Decision (Diamond):** A diamond-shaped symbol representing whether or not you check connectivity.
- **Process (Rectangle):** A rectangle-shaped symbol representing a process, i.e., you are executing the YOLOv8 model.

### 9.3 Detailed Design

The design of the internal structure and functions of each component of the system are all included in the detailed design.

**User Module:** Allows users to create an account, log in, submit their symptoms and upload images of their water sample(s). It provides users with alerts and health recommendations in real-time.

**Data Processing Module:** Receives user information as well as data from the water quality testing process in order to clean and organize that data so that it can be analysed.

**Machine Learning Module:** Studies the user input data and uses it with trained models to identify possible risks associated with disease. The machine learning module will provide the user with a risk score and will identify trends in the data collected from the user.

**Notification Module:** Sends users and local enforcement officials notifications regarding any predicted risk.

**Admin Dashboard:** Provides local law enforcement with the opportunity to monitor user data, trend and report in real-time in order for them to make informed decisions.

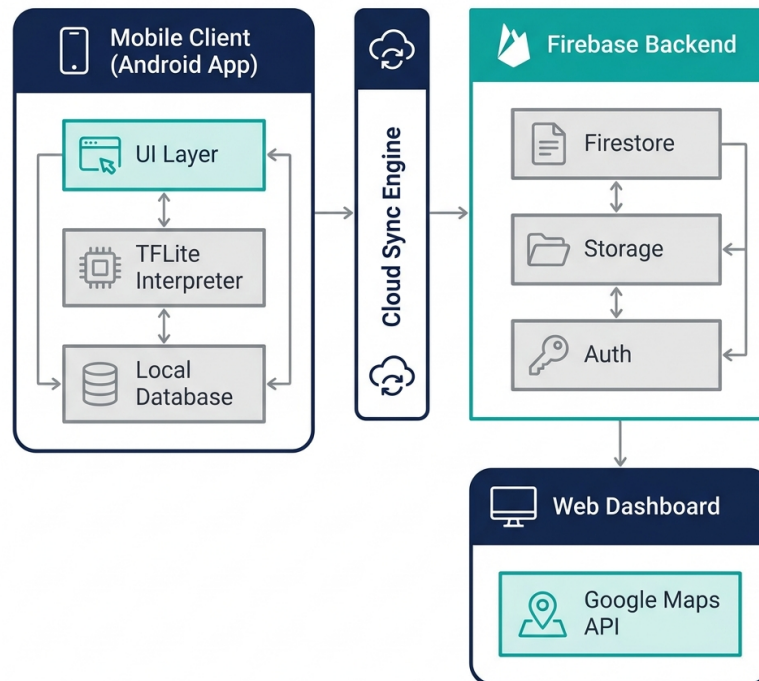
**HIGH-LEVEL SYSTEM ARCHITECTURE: MOBILE HEALTH SYSTEM**

Figure 3: Detailed Design

## 9.4 Flowcharts

A flowchart shows how a system works logically. It shows how the data is being collected, processed, and analyzed in order to produce results.

The flowchart will contain input from the user, data validation, processing of the data. Generating predictions based on the processed data, and generating alerts based on the prediction. By using a flowchart, it is easy to see how the various steps of operation interact and which areas can be improved.

## Mobile App Technical Process Flow

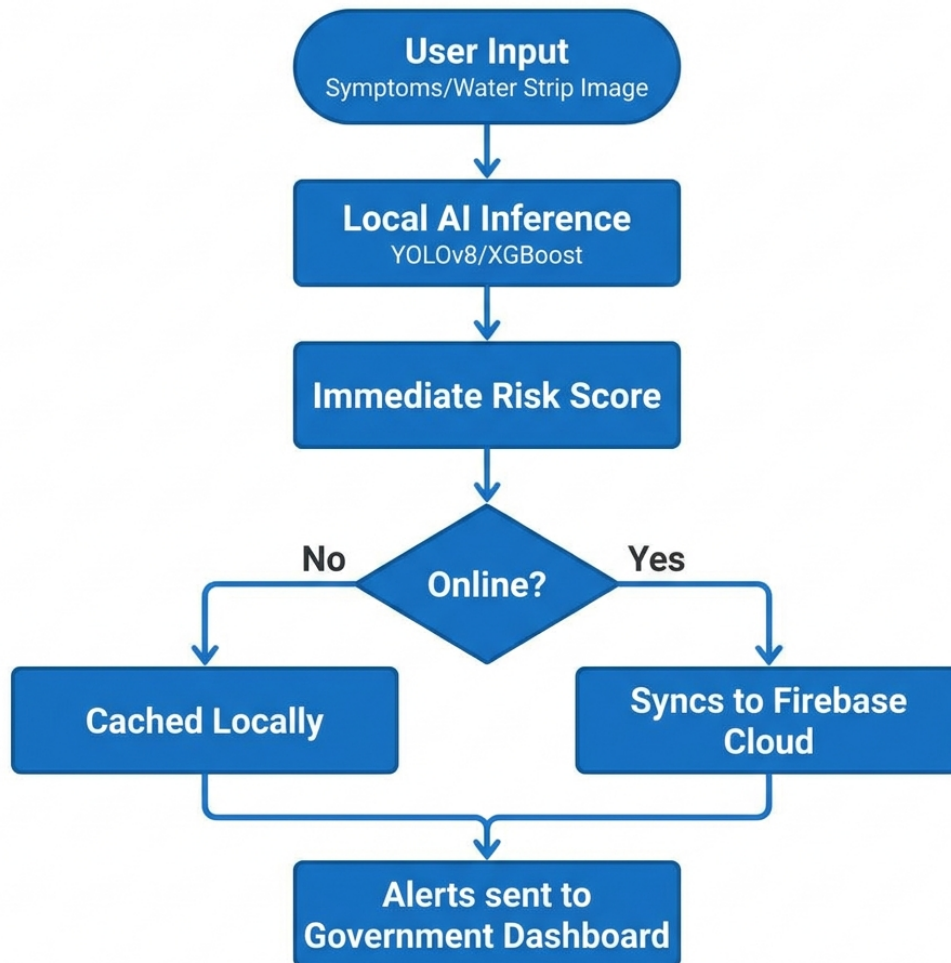


Figure 4: Flowchart of Proposed System

## 9.5 Pseudo Code

### 9.5.1 User Authentication

**Input:** User Credentials (Email, Password), User Type

1. Receive input credentials from the user interface.
2. Validate input format.
  - If invalid, return error and stop.
3. Check User Type.
  - If Government, verify authorized email domain.
  - If invalid, return unauthorized error.
4. Send credentials to authentication server.
5. If authentication is successful:
  - Generate session token.
  - Fetch user profile.
  - Redirect to dashboard based on user type.
6. Else return "Access Denied".

### 9.5.2 Complaint Reporting Module

**Input:** Issue Category, Description, Urgency

1. Request GPS permission.
2. If granted:
  - Capture coordinates and convert to address.
3. Else ask user for manual location.
4. Receive issue details.
5. Validate required fields.
6. Append system data (date, user ID, status).
7. Store data in database.
8. Notify user of success or failure.

### 9.5.3 Water Quality Analysis Module

**Input:** Water Image OR Manual Values

1. Load ML models.
2. If image provided:
  - Detect test strip.
  - Extract color data.
  - Predict chemical values.
3. Read all parameters.
4. Compare with safety thresholds.
5. Send data to server.
6. If safe:
  - Mark as Safe.
7. Else:
  - Mark as Hazardous.
  - Notify government.
8. Store results and display report.

### 9.5.4 Government Analytics Module

**Input:** Citizen Reports and Water Data

1. Connect to database.
2. Initialize counters.
3. Process reports:
  - Count total and critical issues.
4. Process water tests:
  - Identify unsafe results.
  - Map locations.

5. Detect hotspots.
6. Update dashboard.
7. Refresh continuously.

### 9.5.5 Alert Generation Module

**Input:** Hazardous Zones

1. Monitor unsafe zones.
2. If threshold exceeded:
  - Generate warning.
  - Identify affected users.
3. Send notifications.
4. Log alerts.

# 10. TESTING

## 10.1 Functional Testing

Functional Tests are performed to confirm (verify) that the system works as intended, according to the specifications that were set forth in the requirements of the project. Example of the types of things that functional tests focus on testing are the main features of the system, whether or not the system provides the correct output given both valid input and invalid input provided to the system.

Functional testing will also ensure that all of the modules within the proposed AI-Driven Smart Community Health Monitoring System, such as user authentication, data input, image upload, predictive system and the alert generation function enables these features to work as anticipated and perform adequately.

### Test Cases:

| Test Case ID | Module        | Test Description                  | Expected Result                 |
|--------------|---------------|-----------------------------------|---------------------------------|
| TC01         | User Login    | User enters valid credentials     | User successfully logs in       |
| TC02         | User Login    | User enters invalid credentials   | Error message is displayed      |
| TC03         | Symptom Input | User inputs symptoms correctly    | Data is accepted and processed  |
| TC04         | Image Upload  | User uploads water sample image   | Image is uploaded successfully  |
| TC05         | Prediction    | System processes input data       | Disease risk is predicted       |
| TC06         | Alert System  | High risk detected                | Alert notification is generated |
| TC07         | Dashboard     | Admin views system data           | Data is displayed correctly     |
| TC08         | Offline Mode  | User inputs data without internet | Data is stored and synced later |

The above test cases ensure that all functional components of the system are working as intended and meet the specified requirements.

## 10.2 Structural Testing

Structural testing or "white-box" testing helps verify that the internal structure (e.g., code, control flow and data flow) of the system has been implemented correctly.

Each module included in the proposed system (e.g., user authentication, data processing, machine learning prediction and notification handling) will be subjected to structural testing. The purpose of structural testing is to verify that the execution of all logical paths, conditions, and loops occur as expected.

In order to validate the correctness of the implementation, different techniques will be used for structural testing (e.g., paths, conditions and loops). These techniques help in identifying any logical errors, improving code quality and ensuring reliable performance of the system.

## 10.3 Levels of Testing

Software testing takes place at multiple levels in order to confirm that the system does not contain any mistakes and meets all requirements. The levels of testing completed are:

- 1. Unit Testing:** Different components or modules will be tested separately to determine if they are working correctly. An example of this would be a login function, the data input section, or the prediction section of a system.
- 2. Integration Testing:** Different modules will be brought together and tested in combination with each other, in order to confirm they work properly together. An example of this would be to verify the connection between a mobile app and its back-end server.
- 3. System Testing:** The system as a whole will be tested to verify all functions are working correctly together, full work-flow of showing the user input to generating the results will be tested as well.

**4. Acceptance Testing:** A user perspective will be used to test that the system meets the criteria or expectations set forth, including: usability, performance and correctness of the output generated.

## 10.4 Testing the Project

To validate the system's functioning, project testing is done under various conditions, using a variety of test scenarios to check how well the system works for; normal users, edge case users, and error situations. In addition, testing will show validity of input provided by users for; correct data supplied to system, incorrect data provided to system, and incomplete data supplied to the system to assess how robust and reliable the system is; further validation is done to determine how quickly the system responds and can handle the number of users.

The machine learning module will receive significant test coverage to be able to ensure the accuracy of the predictions returned; Further, the alert system will have significant tests to verify that the correct alerts are generated and delivered for cases classified as high-risk.

The overall results of the test activity will demonstrate that the system has been proven to be; dependable, efficient for operation in real-world deployment(s).

## 11. IMPLEMENTATION

### 11.1 Implementation of the Project

The process of implementing the system consists of three main components: creating the application, developing the backend services, and training and deploying the machine learning models to be used in the system.

The application is intended to provide a user-friendly interface for the users of the system to enter symptoms and upload pictures of water samples, as well as receive alerts about water quality. The backend services will utilize cloud-based infrastructure, such as Amazon AWS or Microsoft Azure, to store and process all data collected from each user and communicate with all different components of the system.

The machine learning models developed for this project will be trained to analyze the data entered into the system and predict the likelihood that a person will develop a disease based on the data collected. These trained models will have been trained using data that has been used previously to predict whether a person will develop a disease, and the models will be optimized so that they can produce accurate predictions in a short amount of time. The system will be designed so that users can also enter data into the system without being connected to the internet and can sync the data when they are connected.

Overall, the implementation of the system will function properly, enabling all components to work together (the mobile application, backend services, and machine learning models) to provide the user with real-time updates regarding the status of the water he/she is consuming, as well as predict any potential health-related issues associated with the water.

### 11.2 Conversion Plan

The Conversion Application describes how the existing system will change to a new proposed digital system, through a mostly manual process. The approach to transition from the existing manual processes will be to use a phased approach so that users can phase into using digitized applications.

The initial introduction of the application will be in a few select pilot sites to help test its capabilities and usability. User training and the use of training sessions will help to demonstrate the application and its features to any users and/or decision-makers that will be using the application. It is expected that both manual and automated systems for each of the functions being converted will continue during the transition period to provide users with confidence in the system.

Once the testing and user adoption has been successful, the application will be rolled out to an entire region. Available data on existing records will be converted from paper to electronic formats and uploaded into the converted application. The Conversion Process is designed to facilitate as seamless a transition from existing processes to new digitized processes as possible; however, the conversion process will also provide for as little disruption to existing processing, once the new digitized systems are fully implemented.

### 11.3 Post-Implementation and Software Maintenance

Once the system is launched, ongoing maintenance and monitoring will continue so that the system will perform well into the future with dependable service. Bug fixes, enhancements to overall performance, and regular updates are examples of what would occur after the system has been installed.

By obtaining user feedback, enhancements can be made to system functionality and/or areas of improvement identified. Algorithms behind machine learning models will continue to be re-trained and improved upon based on new data for the most accurate predictions as time progresses. Additionally, ongoing security updates will protect user data as well as ensure the overall integrity of the entire system.

Ongoing technical assistance will be provided to both users and relevant authorities to maintain the reliability and functionality of the entire system while resolving any ongoing issues. Maintenance will continue to keep the entire system reliable, efficient, and able to grow as the requirements of any end users change over a period of time.

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## 12. PROJECT LEGACY

### 12.1 Current Status of the Project

This AI-Driven Smart Community Health Monitoring Early Warning System has been developed and implemented to operate as intended. Core operations for the system have been created, including the ability to collect user inputs, analyze data on the quality of water and predict potential risks of diseases related to the consumption of water by utilizing machine learning techniques.

The application interface for the iOS and Android mobile devices is fully functional and allows users to record symptoms and upload images of their water samples. The backend components of the system are able to efficiently process the information received from the users and generate alerts for users about predicted outcomes based upon their symptoms related to the consumption of water.

The connections between the different components of the system have been thoroughly tested and the system is proven to be highly reliable when operating in normal environmental conditions.

The system has been created and is currently in operation; however, enhancements can be made to increase the accuracy, scalability and usability of the system.

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### 12.2 Remaining Areas of Concern

There are some areas that still need improvement even if implementation has been successful so far. For example, how accurate the machine learning model can be will rely on having good quality training data with lots of different examples; this may be possible to improve by using real-world example from a variety of sources to increase the geographic coverage of the database.

Another issue with the solution is that it relies heavily on mobile devices and user input for its data. Sometimes, this means that the data collected will often be too little or have other issues that make it unusable; this can be a challenge in more remote locations where there is not always reliable mobile service for communication. In addition to this, if a mobile device loses its network connection while submitting data through an online application, that you can still have the

data submitted at a later date once the device reconnects to the internet. However, while there is an initial delay in sending the data, the functionality of online submission without a connection will only work partially without having a system to monitor submissions once the internet becomes available and how that will affect the accuracy of the information received.

Security and privacy of user information is also something that we need to monitor periodically, making sure we have sufficient protections in place and continually working on improvements to the security of user data through the implementation of various security technologies. In addition to monitoring the security/accessibility of user data, the scalability of the solution will also be improved through an evaluation of how the solution can be made to work for a larger scale deployment than what exists today.

### 12.3 Technical and Managerial Lessons Learnt

This project's development helped to provide insight into both technical and managerial aspects of a project. Technically, the project helped to develop an improved understanding of how to develop mobile applications, integrate and utilize a cloud-based system, and implement machine learning methodologies. It also provided hands-on experience working with real-world data, as well as the process of designing and creating systems that can scale as needed.

Managerially, this project pointed out the necessity of proper planning, time allocation and teamwork. There were significant factors that contributed to the successful completion of the project by members of the team. These factors included the level of coordination between members of the team, effective communication and problem-solving skills.

Finally, this project has provided experience that will serve to aid in any future work where the evolving nature of technology has been and will be present.

# USER MANUAL

## Introduction

Sahas 2.0 is a mobile application designed to monitor health risks and water quality using AI. It helps users analyze symptoms, check water safety, and receive alerts.

## Key Features

- Water Quality Analysis
- AI Symptom Checker
- Hospital Locator & Appointment
- Emergency SOS
- Government Alerts

## How to Use

1. Open Water Test
2. Capture test strip image
3. View result (Safe/Unsafe)

## Sympton Checker

1. Enter symptoms
2. Get AI-based advice

## Appointment Booking

1. Enable location
2. Select hospital
3. Book appointment

## Emergency SOS

1. Press SOS button

## 2. Location shared instantly

### Requirements

- Android device (API 24+)
- Internet (optional)
- Camera access

### Troubleshooting

- Camera issue → Ensure proper lighting
- No hospital data → Enable location
- Slow app → Check internet

## 13. SOURCE CODE OR SYSTEM SNAPSHOTS

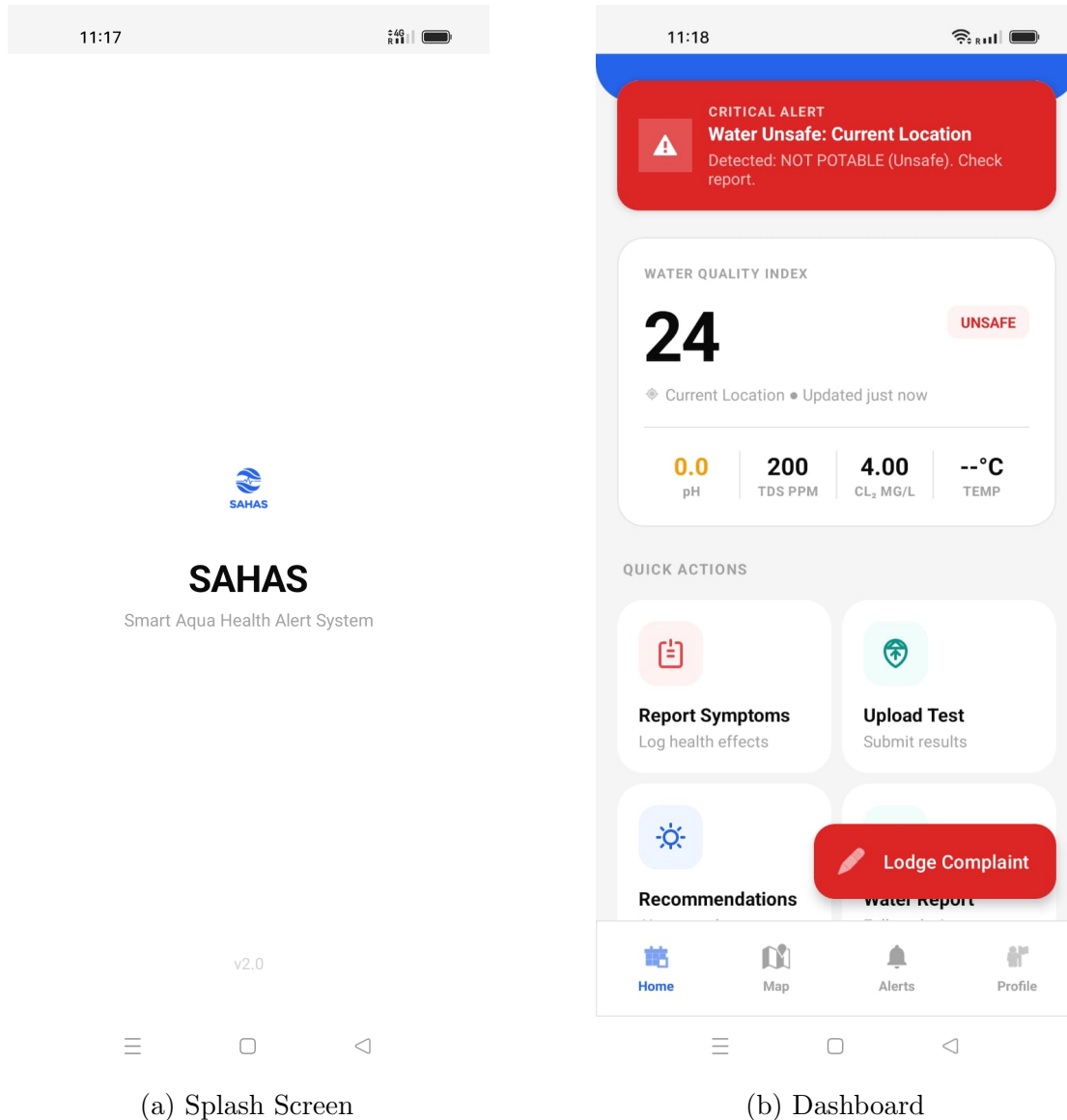
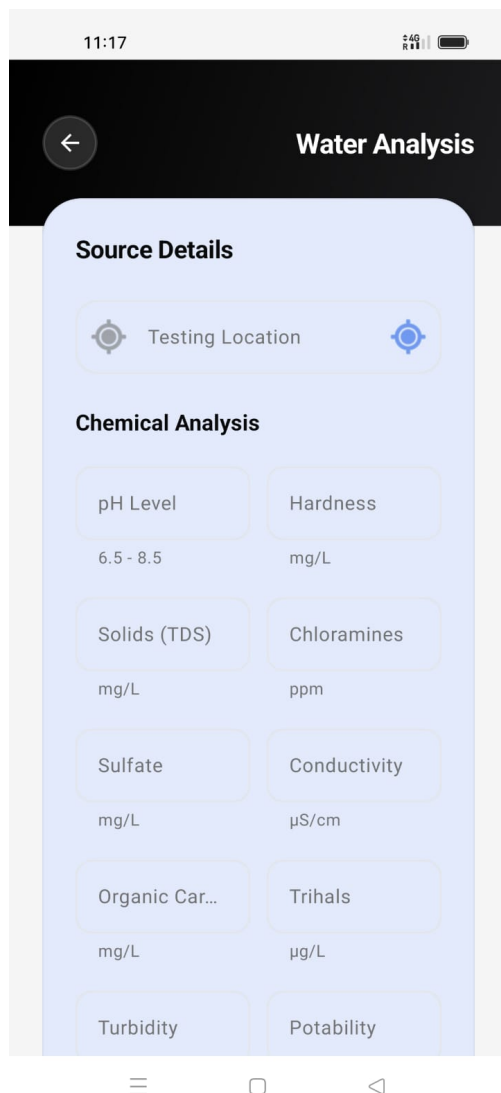
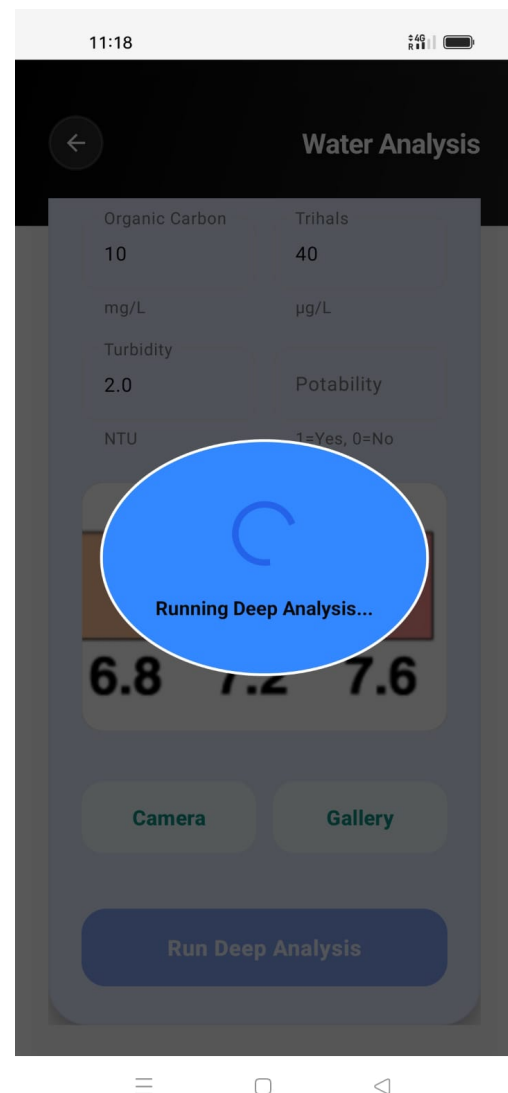


Figure 5: SAHAS Mobile Application Screens (Part 1)



(a) Deep Analysis



(b) Water Input Screen

Figure 6: SAHAS Mobile Application Screens (Part 2)

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